

Solar Subsidies are Booming

Recent developments around the United States are creating a very favorable atmosphere for solar energy. The biggest boost comes from a marked rise in the cost of conventional fuels; by the end of 2005, prices for fuel oil had on average risen 44% nationally in one year, and higher in some regions. Natural gas rose an average of 38% over the same period, though it dropped slightly during the spring of 2006. The reasons for this price increase range from bottlenecks in natural gas production caused by Hurricane Katrina to the global rise in oil prices. These price shocks have raised public worries about energy to levels not seen since the 1970s.

The timing could not be better for the solar-energy subsidies, which have been accumulating from several sources. Tired of waiting for the federal government to act in support of solar, many states have enacted their own tax credits or rebate programs. Each has different regulations and subsidies affecting solar. One state may have subsidies totaling 50%, while an adjacent state may have none at all.

Separately, some state utility regulatory agencies are requiring that a portion of electricity supply come from renewable energy, including solar. Add in a new 30% federal tax credit from the 2005 Energy Policy Act (EPAct), which applies to all states, and in some regions solar seems destined to blossom.

State Incentives

Tax credits are direct reductions in taxes paid by individuals or businesses. Counting sales and property tax exemptions, 15 states offer some form of tax incentive for homeowners. Some are quite generous, at least with commercial building interests: North Carolina offers a 35% tax credit on most renewables, with the maximum amount for commercial systems recently increased to \$2.5 million per installation. For com-

parison, their tax credit for residential solar domestic hot water (DHW) is capped at a paltry \$1,400. Tax credits were the instrument of choice in the

Twenty-four states and municipalities currently offer some form of solar rebate.

California, for example, recently let its tax credit expire, installing in its place



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1970s, but today they are increasingly being displaced by an alternative vehicle: state subsidized rebates.

Tax credits require paying full price and then waiting for up to a year before actually receiving the benefit of a lowered tax bill. Rebates are available immediately after sale. In cases where rebates can be received directly by the solar retailer, the purchase price itself is lower—ideal from a sales perspective. This has resulted in high participation by retailers, with marketing partnerships and public awareness campaigns that typically provide greater visibility for solar programs than do tax credit programs. The funds for these rebates, also known as buy-downs, come mostly from small surcharges on utility bills. These system benefit charges are then spent on measures that reduce electricity and/or natural gas consumption.

a vastly expanded rebate program. A bill setting the goal of one million new PV installations over the next decade failed to pass in the legislature; but with the support of Governor Arnold Schwarzenegger, a program with the same goal was achieved via regulations recently promulgated by the California Public Utilities Commission. Over ten years, \$3.2 billion of funding for this program will be provided.

Renewable Portfolio Standards

Electricity distribution and generation in America is traditionally governed by state regulatory agencies. A number of states are requiring that a minimum percentage of the electric mix be generated by renewables. This is known as a renewable portfolio standard (RPS).

Since regulatory officials are usually appointed rather than elected, regulatory agencies often set more ambitious goals than politicized state legislatures can agree on. Perhaps the best example of legislative gridlock is the recent failure of an effort to get a nationwide RPS included in the massive 2005 EPAct.

based on calculated output. The way thermal kWh are valued for determining payments is based on the electricity output they displace. Predictably, this formula also varies state by state.

Utility solar-electricity buyback programs resemble the German feed-in tariffs, and are fairly generous. Tennessee

to actually buying green electrons. For instance, if a utility wants to meet its RPS requirement with PV, but does not want to own thousands of dispersed PV systems, it can purchase their RECs and in effect subsidize many installations without having responsibility for the hardware. Third-party certification programs ensure proper accounting. RECs can be used for compliance in meeting RPS requirements in nearly all states that have an RPS.

While the concept is not exactly intuitive, the market for RECs is significant and may grow substantially larger. RECs are a valuable tool. Another increasingly popular tool used to help stimulate renewable energy activities is the selling of greenhouse gas emission reduction credits, or offsets. Like RECs, some offsets are purchased on a voluntary basis while others are used for compliance with mandatory emissions limitations in regulatory programs.

RPS standards requiring increases in solar generation have already produced some dramatic results. The biggest PV plant yet announced worldwide, which will produce 18 MW, is pending in Nevada. Arizona Public Service, an electric utility, is offering credits of \$0.07/kWh for solar DHW and an astonishing \$0.16/kWh for solar air conditioning. These payments are just for the renewables contribution credit—the owners of the system are free to sell or use the heat and cooling.

New Jersey has combined an ambitious RPS goal with a significant long-term solar commitment in its public benefits spending. It even has an online trading system for RECs. California also has stable long-term incentives, which, combined with high electricity prices, good net metering rules, and simplified interconnection requirements, bode extremely well for solar growth there. These two states are generally regarded as having the country's leading solar programs, with New York not far behind.



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Of the states that have an RPS, seven specify a minimum percentage for solar. This is significant, since solar usually costs more than other renewable electricity sources, such as wind. Two of these states count only solar PV, but five states specifically allow even solar thermal to be included. These five states are Arizona, Nevada, Pennsylvania, Hawaii, and Texas. (Actually, Texas only requires 500 MW of “nonwind”; it does not require solar.)

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Valley Authority, the largest public utility in the United States, pays \$0.15/kWh for customer-generated solar electricity. We Energies, serving one million customers in Wisconsin, pays \$0.225/kWh. The utility then resells the solar electricity as part of its generation mix. Only the state of Washington uses the feed-in tariff model statewide.

Another factor supporting solar energy use is the market for renewable energy credits (RECs), also known as Green Tags (see “Clean Up Your Ecological Footprint,” *HE* May/June '05, p. 12). RECs represent the environmental attributes of renewable energy generation, which can be unbundled from actual generation and sold separately. This allows the utility to meet mandatory or voluntary goals for renewables contribution by purchasing the REC, as opposed

Federal Incentives

The 2005 EPAct is widely regarded in the renewables community as a triumph for the fossil fuel and nuclear industries, which garnered the vast majority of subsidies. Some significant support for solar energy is also included—with one small catch. There has long been a permanent federal tax credit for commercial solar systems (PV and thermal) of 10%, with no ceiling. This credit has now been raised to 30%. There is no dollar limit on tax credits for commercial installations. A new 30% credit also applies to residential systems, capped at \$ 2,000 for each installation of PV, solar air heating, and solar hot water. A single residence could theoretically receive credits for all three systems. Unfortunately, these new subsidies disappear at the end of 2007. For solar systems that begin operation after that date, the tax credit reverts to 10% commercial and zero residential. The original Senate bill had the 30% credits lasting for six years,

and the political effort now will focus on extending the date beyond next year.

The remaining significant federal solar subsidy is modified accelerated cost recovery system (MACRS), commonly known as accelerated depreciation. Businesses may deduct the entire cost of a solar system from their taxable income over the course of five years. Where applicable, this provides another dramatic subsidy for solar. Like the permanent 10% solar tax credit, MACRS continues after the increased subsidies expire at the end of 2007.

Federal incentives, state solar rebates or tax credits, and utility RPS funds may be available in combination for a single project. For instance, Hawaiians buying a solar water heater could receive \$1,000 from their utility company, a 35% tax credit from the state, and an additional 30% credit on their federal taxes.

The Market for Solar

The variations in subsidy levels are paralleled by dramatic differences among regional solar markets. Outside

Hawaii, California, and Florida, nationwide sales of solar DHW systems have been anemic. Solar combination systems for heating and DHW are very rare. Solar pool-heating systems are more accepted, comprising the biggest U.S. solar thermal market and now totaling over 100,000 installations. PV business is already booming in some areas, such as California, thanks to existing programs. Recently, solar businesses of all kinds report experiencing a dramatic increase in consumer inquiries.



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FOR MORE INFORMATION:

For the Database of State Incentives for Renewable Energy (DSIRE) from the Interstate Renewable Energy Council (IREC) and the North Carolina Solar Center, go to www.dsireusa.org.